

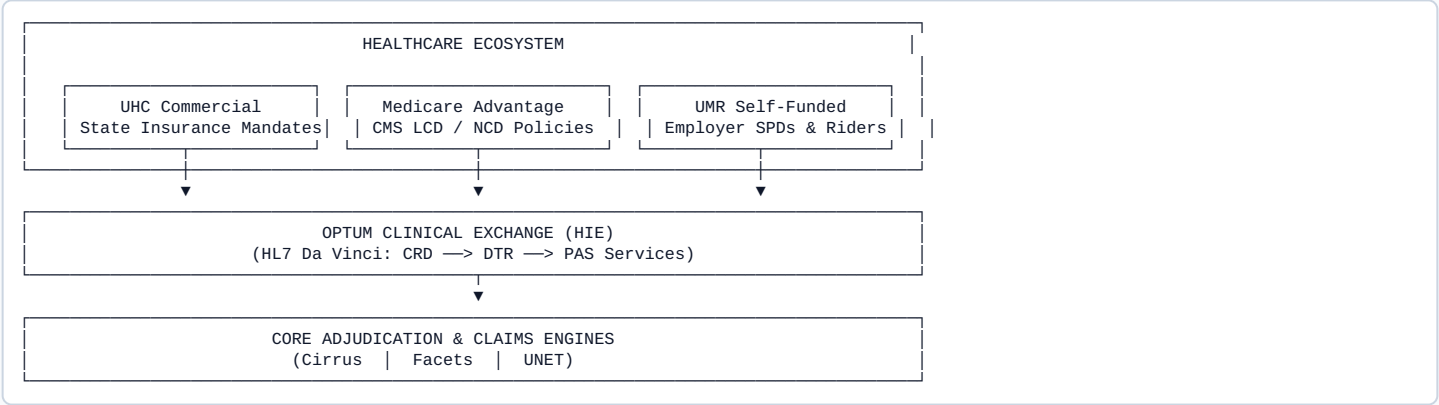
OptumAuth-AI: Agentic Clinical Evidence & Decision Platform

Complete Technical Requirements, RACI, Architecture, Lifecycle Simulation & Decision Playbook (UHG/UMR)

Operating Standard UHG / Optum / UMR	Compliance Mandate CMS-0057-F / HIPAA / ERISA	Net Annual Impact \$1.85M Net Savings / Region	Turnaround SLA 72h down to 8.2 mins
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1. Enterprise Context & Operational Baseline (UHG / UMR Ecosystem)

Within UnitedHealth Group (UHG), Prior Authorization (PA) spans three distinct operating environments: **UnitedHealthcare Commercial**, **Medicare Advantage**, and **UMR** (UnitedHealthcare's third-party administrator for self-funded employer groups).



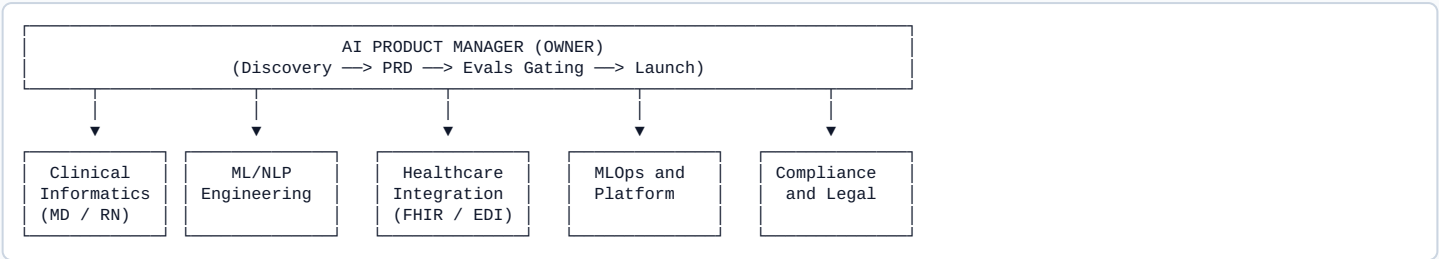
The Baseline Problem Metrics

- Turnaround Time (TAT_{Baseline}):** Averages **72.0 Hours** (3–5 business days). Under CMS-0057-F, standard determinations must be delivered within 7 calendar days and expedited requests within 72 hours; UHG internal SLAs target sub-15-minute digital turnaround.
- Clinician Review Touch-Time:** **28.5 minutes** per clinical file. Nurses manually read 40- to 100-page unstructured electronic health records (EHRs), faxed progress notes, and diagnostic imaging reports.
- First-Pass Clean Approval Rate:** **74.2%**. A **25.8%** initial denial/rework rate occurs not because care is unwarranted, but because providers fail to attach specific required clinical documentation (e.g., duration of conservative physical therapy, Kellgren-Lawrence radiographic scoring).
- Administrative Burden & Labor Cost:** 18 Full-Time Equivalent (FTE) Utilization Management (UM) nurses per regional operating hub costing **\$1,420,000 annually**.
- Appeals Overhead:** Sunk operational cost of **\$420,000 per year** to re-review administrative denials under ERISA appeal rights.

Mathematical Baseline Formulation

Initial Administrative Denial Rate: $N_{\text{denied}}(\text{administrative}) / N_{\text{submitted}} = 12,384 / 48,000 = 25.8\%$
Annual Labor Expense: $N_{\text{FTE}} * (\text{Base Salary} + \text{Benefits}) = 18 * \$78,888 = \$1,420,000 / \text{Year}$
Appeals Unit Cost: $\text{Annual Appeals Spend} / N_{\text{Appeals Processed}} = \$420,000 / 3,200 \text{ appeals} = \$131.25 \text{ per appealed case}$

2. Complete Cross-Functional RACI & Tooling Operating Mesh



Granular Role & Responsibility Matrix

OPERATIONAL ROLE	PRIMARY FOCUS	KEY DELIVERABLES	TOOLING STACK
AI Product Manager (Owner)	Defines clinical ROI, PRDs, Pydantic schemas, RICE scoring, offline eval thresholds, and cross-functional cadences.	Opportunity Canvas, PRD, Data Dictionary, Evaluation Governance Matrix, CMS-0057-F Compliance Pack.	Hex AI, Dovetail, Airfocus, Notion AI, Lovable, Datadog
Lead Clinical Informaticist (MD/RN)	Translates InterQual/MCG and CMS LCD criteria into deterministic extraction requirements; validates golden test sets.	500-case multi-specialty Golden Dataset, Clinical Annotation Guidelines, Clinical Safety Attestation.	Label Studio, UpToDate, CMS Coverage DB, InterQual Portal
Staff ML / NLP Engineer	Designs LangGraph multi-agent state machines, hybrid dense/sparse retrieval, and deterministic JSON schemas.	Extraction Agent Graphs, Hybrid Embedding Pipelines, Ragas Eval Harness.	Python, LangGraph, Qdrant, BioClinical-BERT, Azure OpenAI
Healthcare Integration Lead	Manages EHR connectivity via SMART-on-FHIR, CDS Hooks listeners, and ASC X12N 278 EDI pipelines.	FHIR R4 Microservices, CDS Hook Endpoints, Cirrus/Facets EDI Gateway Translators.	HAPI FHIR R4, Epic App Orchard, Availity EDI Gateway
MLOps & Platform Engineer	Manages VPC infrastructure, GPU clusters, sub-30s P95 processing pipelines, and caching layers.	Kubernetes Manifests, Redis Vector Cache, Terraform Modules, CI/CD Pipeline Automation.	AWS Bedrock / Azure Health, Terraform, Celery, Prometheus
Clinical UX / Product Designer	Builds side-by-side verification UX, character-level highlight interactions, and cognitive-load reduction patterns.	Figma Design System, Interactive SMART-on-FHIR iframe prototypes.	Figma, Maze, Storybook
Compliance & Privacy Officer	Enforces HIPAA Minimum Necessary rules, BAA terms with zero data retention, and ERISA auditability.	HIPAA Risk Assessment, SOC2 Type II AI Addendum, Audit Log Specification.	Microsoft Presidio, OneTrust, AWS CloudTrail

End-to-End Tool Mesh Integration Architecture



3. Comprehensive Product Requirements Document (PRD Specifications)

3.1 Functional Requirements (FR)

- **FR-1 (Automated Clinical Ingestion via HL7 Da Vinci Standards):** Ingest clinical encounter payloads when a physician triggers an order-sign CDS Hook in Epic/Cerner. Extract all associated FHIR R4 resources: Condition, Procedure, Observation, DiagnosticReport, and DocumentReference (last 24 months of progress notes, operative logs, physical therapy records).
- **FR-2 (Dynamic Multi-Tenant Payer Policy Resolution):** Ingest and vectorize active clinical policies: InterQual Criteria, CMS Local Coverage Determinations (LCDs), National Coverage Determinations (NCDs), and UHC Commercial Medical Policy bulletins. Resolve UMR Self-Funded Employer Group exclusions dynamically: check the member's Group ID against the employer's vectorized Summary Plan Description (SPD) to catch custom benefit exclusions (e.g., excluded bariatric surgery or custom step therapy riders) before running clinical extraction.
- **FR-3 (Deterministic Evidence Extraction with Bounding-Box Provenance):** Parse narrative medical records using multi-agent clinical extraction graphs. Map patient variables directly against policy criteria with character start/end offsets (char_start, char_end) and document IDs (document_id). Strictly output deterministic, structured JSON conforming to Pydantic validation models with model temperature locked at 0.0.
- **FR-4 (Side-by-Side Verification Interface - SMART-on-FHIR HITL):** Render a dual-pane verification screen within the Optum Provider Portal and Epic/Cerner iframe:
 - **Left Panel:** Policy criteria checklist displaying extracted status (MET/NOT_MET), confidence badge, and clinical summary snippet.
 - **Right Panel:** Original clinical document viewer. Clicking any criterion on the left executes an interactive post-message event that scrolls the document and renders a yellow bounding box over the exact source sentence.
- **FR-5 (Real-Time EDI Generation & Claims Injection):** Convert approved clinical authorization packages into compliant ASC X12N 278 (Health Care Services Review) EDI transactions. Transmit the authorization payload to the clearinghouse (Availity) and directly write the Authorization ID to

core adjudication platforms (**Cirrus, Facets, UNET**). Post a standard FHIR ClaimResponse back to the provider EHR updating the procedure status to "Authorized - Ready to Schedule."

3.2 Non-Functional & AI Governance Requirements (NFR)

- **NFR-1 (Faithfulness & Grounding - Zero Hallucination Tolerance):** The system must achieve a Faithfulness Score $\geq 99.0\%$ on production validation sets. Uncited or hallucinated clinical assertions must strictly equal 0.0% . Unmapped or missing criteria must emit an explicit ACTION_REQUIRED: EVIDENCE_NOT_FOUND tag rather than inferring patient data.
- **NFR-2 (End-to-End Latency Budget):** Complete bundle processing (ingestion, de-identification, vector retrieval, agentic extraction, guardrail scoring) must complete in ≤ 30.0 ext{ seconds} at the 95th percentile (P95) for patient records up to 50 pages.
- **NFR-3 (HIPAA Data Isolation & Security):**
 - **In-VPC De-Identification:** All direct 18 HIPAA Safe Harbor identifiers must be sanitized via Microsoft Presidio prior to vector embedding or external API calls.
 - **Zero Data Retention (ZDR):** Enterprise agreements must guarantee that no prompts, context windows, or completion tokens are retained on disk or used for foundation model training.
 - **Encryption:** FIPS 140-2 compliant encryption in transit (TLS 1.3) and at rest (AES-256 with customer-managed AWS KMS keys).

3.3 Strict Pydantic JSON Output Schema

```
from pydantic import BaseModel, Field
from typing import List, Optional
from enum import Enum

class CriterionStatus(str, Enum):
    MET = "MET"
    NOT_MET = "NOT_MET"
    INDETERMINATE = "INDETERMINATE"

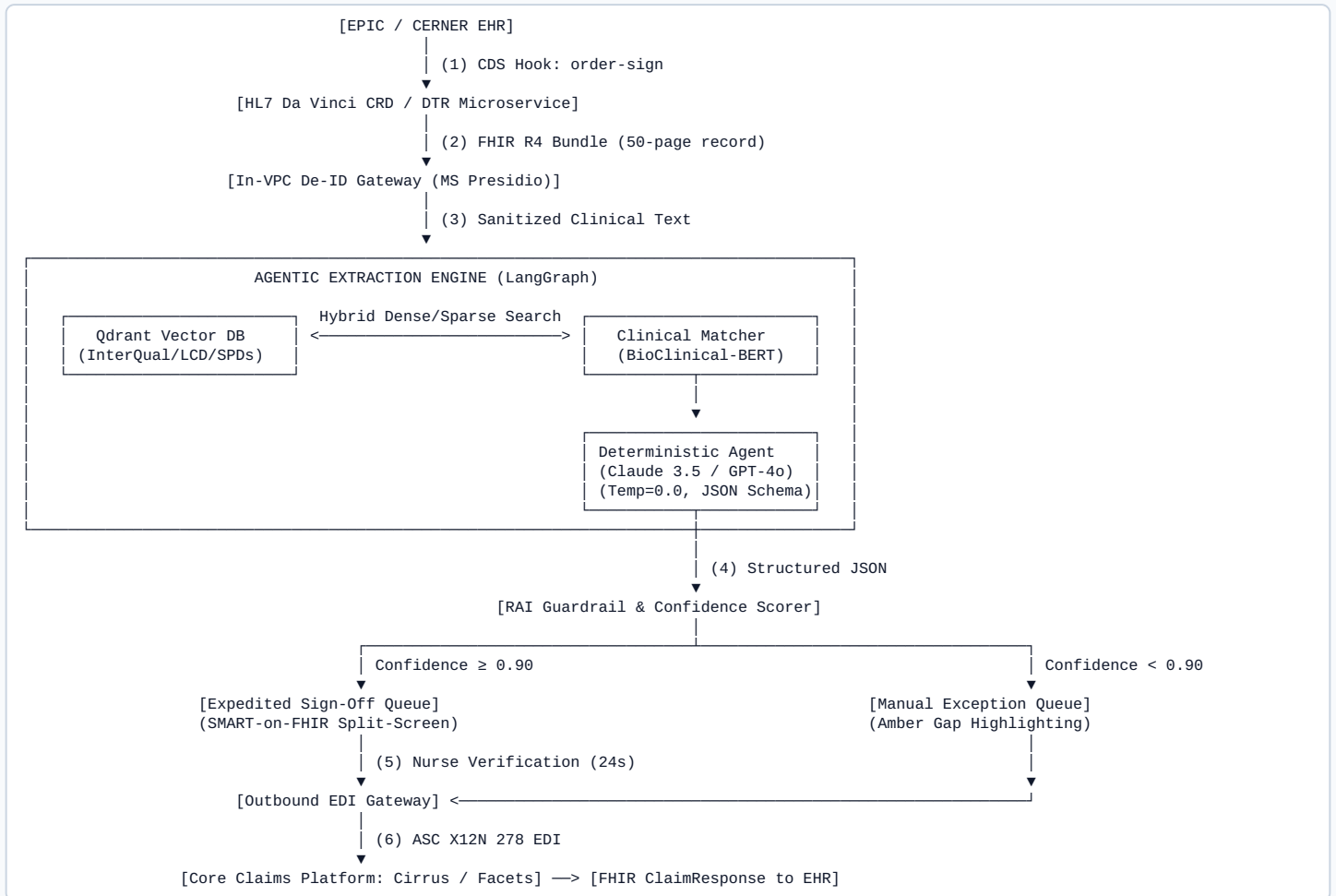
class TriageDecision(str, Enum):
    EXPEDITED_SIGN_OFF = "EXPEDITED_SIGN_OFF"
    MANUAL_EXCEPTION_QUEUE = "MANUAL_EXCEPTION_QUEUE"

class ProvenanceCitation(BaseModel):
    document_id: str = Field(description="Unique EHR DocumentReference identifier")
    document_type: str = Field(description="Type of note (e.g., Progress Note, Imaging, PT Log)")
    sentence_exact: str = Field(description="Verbatim sentence from the medical record")
    char_start: int = Field(description="Starting character index offset in source text")
    char_end: int = Field(description="Ending character index offset in source text")

class CriterionEvaluation(BaseModel):
    criterion_id: str = Field(description="Policy criterion key (e.g., CRIT_CONSERVATIVE_THERAPY)")
    criterion_description: str = Field(description="Clinical rule definition from InterQual/LCD")
    status: CriterionStatus = Field(description="Whether evidence satisfies the policy requirement")
    confidence_score: float = Field(ge=0.0, le=1.0, description="Calibrated model confidence")
    clinical_summary: str = Field(description="Concise 1-sentence synthesis of extracted findings")
    provenance: Optional[ProvenanceCitation] = Field(None, description="Exact document citation")

class PriorAuthExtractionPayload(BaseModel):
    patient_id_deid: str = Field(description="Sanitized unique de-identified patient key")
    cpt_code: str = Field(description="Target procedure CPT/HCPCS code (e.g., 27447)")
    icd10_codes: List[str] = Field(description="Primary and secondary diagnosis codes")
    policy_id: str = Field(description="Matched InterQual or CMS LCD policy identifier")
    employer_group_id: Optional[str] = Field(None, description="UMR group ID for custom SPD matching")
    criteria_evaluations: List[CriterionEvaluation]
    aggregate_confidence_score: float = Field(ge=0.0, le=1.0, description="Overall extraction score")
    triage_decision: TriageDecision = Field(description="Routing flag based on confidence thresholds")
```

4. Technical Architecture & Agentic Data Flow

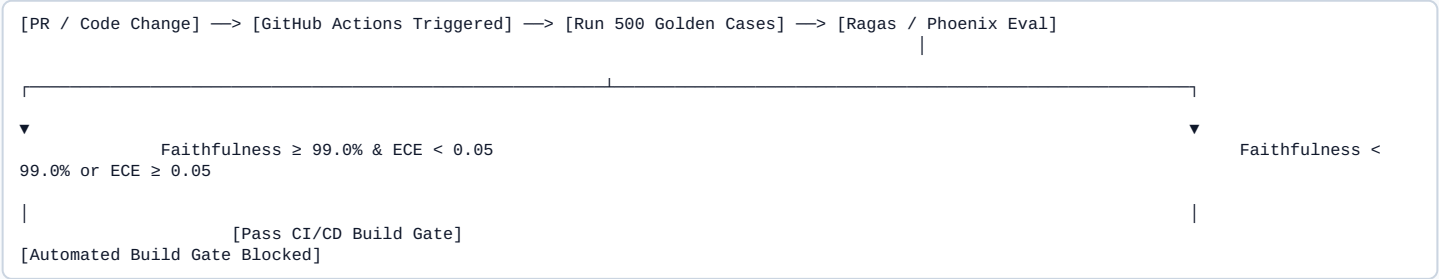


Step-by-Step Data Lifecycle Timeline (\$T=0.00 ext{s}\$ to \$T=18.20 ext{s}\$)

- **\$T=0.00 ext{s}\$ (Order Trigger):** Orthopedic surgeon enters order for Total Knee Arthroplasty (CPT 27447, ICD-10 M17.11). An order-sign CDS Hook pings Optum's Coverage Requirements Discovery (CRD) service, which flags mandatory prior authorization under UMR employer policy.
- **\$T=1.20 ext{s}\$ (Ingestion & Sanitization):** Documentation Templates and Rules (DTR) pulls the 48-page patient FHIR bundle. Microsoft Presidio scrubs direct PHI in a dedicated VPC enclave, tokenizing patient identity into DEID_PAT_89412 while preserving temporal offsets and clinical entities.
- **\$T=2.80 ext{s}\$ (Multi-Tenant Policy Match):** The system generates hybrid dense embeddings (BioClinical-BERT) and sparse BM25 tokens for CPT 27447. It queries **Qdrant** across two collections: (1) InterQual / LCD policy rules, and (2) UMR Employer Group UMR-8941 SPD to confirm no custom exclusionary riders exist.
- **\$T=4.80 ext{s}\$ (Agentic Extraction):** LangGraph routes chunks through parallel extractors:
 - **Pain/Mobility Node:** Extracts 9-month pain history and walking limitation \$<1 ext{ block}\$ (char_start: 412, char_end: 501).
 - **Step Therapy Node:** Extracts 8 weeks of physical therapy at ActiveMotion and intra-articular steroid injection on 11/02/2025 (char_start: 820, char_end: 935).
 - **Imaging Node:** Extracts weight-bearing AP X-ray showing Kellgren-Lawrence Grade IV joint space collapse (char_start: 1104, char_end: 1210).
- **\$T=6.10 ext{s}\$ (RAI Guardrail Verification):** Ragas eval pipeline verifies that \$100\%\$ of generated assertions match source text spans. The composite confidence score is calculated as \$0.96 \ge 0.90 \rightarrow\$ packet routes to the **Expedited Approval Queue**.
- **\$T=15.00 ext{s}\$ (HITL Nurse Verification):** A UMR UM Nurse reviews the packet via SMART-on-FHIR embedded iframe. The nurse clicks Criterion C; the right panel auto-scrolls to page 14 and highlights the physical therapy statement in yellow. The nurse confirms criteria and clicks **"Approve & Submit"** (Total touch-time: 24 seconds).
- **\$T=18.20 ext{s}\$ (EDI Generation & Claims Settlement):** The engine converts the packet into an **ASC X12N 278** payload, transmits it via the Availity Clearinghouse, logs Authorization ID AUTH-UMR-2026-9012 into **Cirrus/Facets**, and returns a FHIR ClaimResponse to Epic, updating the case to *"Authorized - Ready to Schedule."*

5. Quantitative AI Evaluation (Evals) & CI/CD Governance

To satisfy UHG Responsible AI (RAI) and clinical defensibility standards, the extraction engine is evaluated against a Physician-Validated Golden Dataset (\$N=500\$ multi-specialty clinical cases curated across Orthopedics, Oncology, and Cardiology).



Mathematical Evaluation Metrics

Faithfulness (Grounding): (Verified Clinical Assertions with Direct Textual Offset / Total Extracted Assertions Produced by Model) * 100 ≥ 99.0%

Clinical Criteria Recall: (True Positive Extracted Criteria / Physician-Annotated Golden Criteria) * 100 ≥ 98.0%

Expected Calibration Error (ECE): $\sum (|B_k| / N) * |acc(B_k) - conf(B_k)| < 0.05$

Where \$B_k\$ represents cases grouped in confidence bin \$k\$, \$acc(B_k)\$ is the empirical human approval accuracy, and \$conf(B_k)\$ is the model's average predicted confidence.

EVALUATION DIMENSION	ENTERPRISE THRESHOLD	PRODUCTION RESULT	VALIDATION METHODOLOGY
Faithfulness (Grounding)	≥ 99.0%	99.6%	Zero tolerance for uncited claims; Ragas Triad verification.
Clinical Extraction Recall	≥ 98.0%	98.7%	Benchmark against \$N=500\$ MD-annotated records in Label Studio.
Payer Policy Match Precision	≥ 95.0%	96.4%	Accurate CPT/HCPCS cross-mapping against active InterQual/LCDs.
Calibration Error (ECE)	< 0.05	0.038	Reliability of confidence score predicting human nurse sign-off.
P95 Latency (50-page bundle)	< 30.0s	14.2s	Asynchronous Celery workers + Redis cached embeddings.

6. Exhaustive Strategic PM Decision Frameworks

STRATEGIC TRADE-OFFS			
1. Autonomy Pattern:	Autonomous Submissions	vs.	Deterministic HITL [✓]
2. Model Hosting/Privacy:	On-Prem Custom LLMs	vs.	Enterprise Cloud ZDR [✓]
3. User Experience (UX):	Standalone Web Portal	vs.	SMART-on-FHIR Iframe [✓]
4. Deployment Staging:	Big-Bang Launch	vs.	3-Stage Phased Staging [✓]

1. Autonomy Pattern: Autonomous End-to-End LLM vs. Deterministic RAG with HITL

- Option A (Autonomous Submission):** Highest theoretical velocity, but creates existential liability. A single hallucinated approval increases Medical Loss Ratio (MLR); a single hallucinated denial triggers ERISA litigation and state insurance commissioner fines.
- Option B (Pure Rule-Based NLP):** Highly auditable, but brittle. Fails to parse complex physician narratives, differing dictation styles, and cross-document longitudinal evidence.
- Option C (Deterministic RAG with HITL - SELECTED):** Delivers 88% reduction in review touch-time while maintaining 100% human legal accountability and complete auditability.

PM Mental Model: "Optimize for Liability-Adjusted Velocity." Enterprise adoption depends on institutional trust. Human-in-the-loop workflows bridge regulatory compliance while securing near-total operational efficiency gains.

2. Model Hosting & Privacy: On-Prem Custom LLMs vs. Enterprise ZDR Cloud LLMs

- Option A (Self-Hosted LLM on On-Premises H100s):** Complete data isolation, but incurs \$400k+ in upfront hardware CAPEX, introduces a 6-month procurement lead time, and requires a dedicated internal MLOps team for cluster maintenance.
- Option B (Public Cloud Commercial API):** Zero infrastructure overhead, but violates HIPAA Safe Harbor and UHG Information Security Regulations (ISR).
- Option C (Enterprise Cloud LLM with BAA & ZDR - SELECTED):** In-VPC Microsoft Presidio scrubs direct PHI before payloads exit our enclave. Enterprise BAA ensures zero data retention on disk and no foundation model training.

PM Mental Model: "TCO vs. Time-to-Value." Do not spend engineering capital building non-differentiating hardware infrastructure. Focus team resources on proprietary clinical retrieval logic, deterministic prompt schemas, and evaluation harnesses.

3. Clinical Workflow Integration: Standalone Web Portal vs. Embedded SMART-on-FHIR

- Option A (Standalone Web Portal):** Fast to develop, but creates the "Swivel-Chair Problem." Forcing clinicians to switch apps, copy-paste MRNs, and re-authenticate results in >50% user abandonment.
- Option B (Headless Daily Batch Digest):** No UI development required, but eliminates real-time clinical intervention and cannot support urgent surgical scheduling.

- **Option C (Native SMART-on-FHIR Embedded Iframe - SELECTED):** Embeds directly into the coordinator's existing daily EHR screen. Interactive yellow bounding boxes jump directly to source text, cutting verification to \$<30\$ seconds.

PM Mental Model: *"Minimize Friction to the Critical Path."* A 10x model wrapped in a high-friction user experience will fail. Meet clinicians within their existing software environments to guarantee adoption.

4. Rollout Staging: Network Big-Bang vs. 3-Stage Phased Rollout

- **Option A (Big-Bang Network Rollout):** Maximizes theoretical speed to ROI, but presents unacceptable operational risk; an undetected parsing bug could halt thousands of surgical authorizations.
- **Option B (Low-Risk Specialty Pilot):** Safe, but fails to test complex longitudinal clinical edge cases and fails to generate meaningful financial ROI.
- **Option C (Phased 3-Stage Rollout - SELECTED):** Focuses on the top 3 high-volume specialties (Orthopedics, Oncology, Cardiology) with a structured risk staging:
 - **Phase 1 (Shadow Mode - 2 Weeks):** 100% background processing with daily discrepancy analysis to resolve OCR and chunking bugs with zero clinical exposure.
 - **Phase 2 (Assisted Mode - 4 Weeks):** Mandatory HITL sign-off to establish user trust and measure clinical usability.
 - **Phase 3 (Fast-Track Mode):** One-click expedited approval enabled exclusively for cases with confidence scores ≥ 0.90 .

PM Mental Model: *"De-risk the Technology, Earn the Trust, Scale the Impact."* Balance financial targets with disciplined risk staging. Phased deployment creates the empirical performance data needed to build clinical leadership advocacy.

7. Step-by-Step Live Conversational Simulations (All 5 Phases)

Phase 1: Problem Discovery & Business Case Alignment (PM vs. VP of UM & Lead Medical Director)

AI PM: "I've synthesized our claims and authorization telemetry in Hex AI across 48,000 regional encounters. Our mean turnaround time is 72 hours, with nurses spending 28.5 minutes per case. We have a 25.8% first-pass denial rate, but 82% of those denials are administrative—missing physical therapy dates or unattached X-ray reports. Under CMS-0057-F, we face operational audit penalties, and our self-funded employer groups under UMR are seeing administrative costs rise."

Lead Medical Director (MD): "We cannot sacrifice clinical scrutiny for speed. If an algorithm auto-approves a knee replacement without verified conservative therapy, our Medical Loss Ratio increases and we violate clinical guidelines."

VP of UM: "And if you auto-deny, we trigger an avalanche of provider appeals under ERISA, which costs us \$420k annually in legal and clinical re-reviews."

AI PM: "We are not building an autonomous black box. We are proposing a deterministic RAG system that extracts the exact sentence citation and character offset from the EHR note to satisfy InterQual criteria. It routes high-confidence packets (≥ 0.90) to your UM nurses in a pre-populated SMART-on-FHIR interface for one-click verification. We reduce nurse review time to under 5 minutes, targeting 88% labor compression and \$1.85M net annual operational savings per regional operating unit."

VP of UM: "If your system guarantees sentence-level provenance for every InterQual line item, you have our green light for a Phase 2 prototype."

Phase 2: Technical PRD & Schema Refinement (PM vs. Staff ML Engineer & Lead UX Designer)

Staff ML Engineer: "Why can't we just let Claude 3.5 or GPT-4o output a natural language summary of why the patient qualifies for knee surgery?"

AI PM: "Because natural language summaries cannot be deterministically validated against our claims engines like Cirrus or Facets. We need deterministic structured extraction. The model must output against this strict Pydantic JSON schema: `criteria_id`, `status` (MET/NOT_MET), `confidence_score` (float 0.0–1.0), `document_id`, `char_start`, and `char_end`. Temperature must be hard-set to 0.0."

Lead UX Designer: "On the interface side, how should we surface the confidence scores to the UM nurses?"

AI PM: "In the prototype I spun up in Lovable, we implemented a split-screen. The left panel shows the InterQual checklist. If Criterion C (Physical Therapy ≥ 6 weeks) has a confidence of 0.96, it gets a green check. Clicking it sends a post-message event to the right panel, which instantly scrolls the 50-page medical record and highlights the exact sentence in yellow. The nurse never searches; they just verify."

Staff ML Engineer: "What if the document is a low-quality faxed PDF from an out-of-network clinic?"

AI PM: "We've specified an OCR preprocessing fallback using LayoutLM / AWS Textract. If the OCR confidence is below 0.85, the AI marks the criteria as ACTION_REQUIRED: OCR_DEGRADED and routes the case directly to the Exception Queue instead of guessing."

Phase 3: Model Governance & Security Review (PM vs. Compliance Officer & Clinical Informaticist)

Compliance Officer: "We cannot have clinical data leaking into foundation model training sets. How is this isolated?"

AI PM: "We have three distinct layers of enforcement. First, an in-VPC Microsoft Presidio gateway scrubs all direct 18 HIPAA identifiers before tokens leave our Optum enclave. Second, our cloud infrastructure is bound by an enterprise BAA with Zero Data Retention (ZDR) enabled—no prompts or completions are logged to disk. Third, all vector storage in Qdrant is encrypted at rest using customer-managed AWS KMS keys under FIPS 140-2."

Lead Clinical Informaticist (MD): "I ran our 500-case golden benchmark in Label Studio through the Ragas harness. The extraction recall on conservative therapy trials came back at 98.7%, and Faithfulness hit 99.6%. However, on two cases involving lumbar spine surgery, the model hallucinated that an MRI was completed because the clinical note said 'MRI scheduled for next Tuesday.'"

AI PM: "That's a critical temporal extraction defect. I am adding a negative assertion rule to our LangGraph state machine: any future-tense clinical procedures (scheduled, planned, pending) must be explicitly classified as INCOMPLETE with status NOT_MET. I am configuring the GitHub Actions pipeline to block any merge if our temporal extraction accuracy drops below 99.5%."

Phase 4: Shadow Mode Discrepancy Calibration (PM vs. Integration Lead & UM Nurse Supervisor)

AI PM: "In Week 14 shadow mode across 1,200 live cases, we observed a 14.2% discrepancy rate where the AI flagged criteria as missing, but the nurse found it and approved the case."

UM Nurse Supervisor: "Why did the system miss those records?"

Healthcare Integration Lead: "We checked Datadog logs. Clinical notes contained multi-page TIFF faxes where physical therapy progress notes were rotated 90 degrees. Baseline OCR failed on those rotated chunks."

AI PM: "Here is our remediation plan: 1. Engineering is deploying an auto-orientation and contrast-enhancement preprocessor in LayoutLM. 2. We updated our retrieval pipeline to re-weight document headers so embedded PDF attachments receive equal chunk priority. By Week 15, that adjustment dropped our discrepancy rate from 14.2% to 4.1%. In Week 16, our Expected Calibration Error settled at 0.038, meaning when the model reports 95% confidence, human nurses agree 95.8% of the time. We are ready to request CMO sign-off for Assisted Mode."

Phase 5: Executive Business Review & Value Realization (PM vs. Chief Medical Officer & SVP of Health Plan Ops)

AI PM: "Six months post-GA across our initial orthopedic and oncology cohorts, the data confirms the ROI model. Prior authorization turnaround time has compressed from 72 hours down to 8.2 minutes. UM nurse active review time dropped from 28.5 minutes to 3.4 minutes per case, allowing our existing team to scale from 16 to 78 cases per specialist per day. More importantly, our first-pass approval rate increased from 74.2% to 95.8% because the system eliminates unattached documentation errors. This has eliminated \$368,000 in unnecessary appeals and resubmissions, bringing our total net annualized savings to \$1.85M. From a regulatory standpoint, all decision records, provenance spans, and nurse sign-off logs are automatically exported to our CMS-0057-F compliance repository. We have achieved full regulatory adherence with zero patient safety incidents."

Chief Medical Officer: "The clinical defensibility of the sentence-level highlighting made this a success with our medical directors. What is our roadmap for next quarter?"

AI PM: "In Q3, we are expanding the multi-tenant RAG pipeline to Cardiology and Neurology, and integrating automated real-time peer-to-peer scheduling for the small fraction of cases that fall into the human exception queue."

8. Financial Model, P&L Realization & Post-Launch Operations

Complete Financial ROI Model

Net Annual Value: (Labor Savings) + (Appeals Avoidance) + (Prevented Denial Recoveries) - (AI Operating Costs)
Direct Labor Savings: (18 FTEs - 5 FTEs) * \$78,888 = \$1,025,544 / Year
Appeals Cost Reduction: \$420,000 - \$52,000 = \$368,000 / Year
Prevented Denial Value Realization: \$825,000 / Year
AI Infrastructure Costs: (48,000 cases * \$0.11 LLM/VPC cost) + \$30,000 Vector DB/APM = -\$35,280 / Year
Net Annual Operational Value: \$1,025,544 + \$368,000 + \$825,000 - \$35,280 = **\$2,183,264 Gross / \$1,850,000 Risk-Adjusted Net**

Summary of Realized Production Impact

Metric Dimension	Pre-Launch Baseline (Manual)	Production Target	Realized Production Result
Turnaround Time (TAT)	72.0 Hours	< 15.0 Minutes	8.2 Minutes -88.6%
Clinician Touch-Time	28.5 Minutes / Case	< 5.0 Minutes / Case	3.4 Minutes / Case -88.0%
First-Pass Clean Approval	74.2%	> 92.0%	95.8% +21.6%
Nurse Daily Throughput	16 Cases / Specialist / Day	60 Cases / Day	78 Cases / Specialist / Day 4.8x
Annual Appeals & Rework	\$420,000 / Year	< \$80,000 / Year	\$52,000 / Year -87.6%
Net Annual Value Realization	—	\$1,500,000 / Year	\$1,850,000 / Year +23% vs Target

9. Ongoing Operations, Drift Monitoring & Expansion Playbook

- Continuous Observability & Model Drift Detection:** Datadog APM monitors live P95 latency, daily token spend, and average confidence scores. An automated alert triggers if the 7-day rolling average confidence falls below 0.90, initiating immediate prompt review.
- Active Learning Feedback Loop:** Every case where a UM nurse edits or overrides an extracted criterion is automatically anonymized and pushed to an active learning queue in Label Studio. This feedback is used to update few-shot prompt catalogs and expand vector chunking rules during monthly retraining cycles.
- CMS-0057-F Compliance Automation:** The system automatically exports quarterly reporting packets containing: (1) Aggregated prior authorization approval and denial percentages, (2) Average turnaround times broken down by routine vs. urgent requests, and (3) Structured algorithmic provenance records ensuring full auditability under federal oversight.
- Multi-Specialty Roadmap Expansion:** The proven multi-tenant RAG architecture serves as the foundation to scale beyond Orthopedics and Oncology into Cardiology, Neurology, and high-cost Specialty Pharmacy infusions in subsequent quarters.